POMDPs: beliefs and decisions

Bayesian updates and the value of gathering information.

Estimate what is hidden

State and observation
Bayesian belief updates

Decide with uncertainty

Information value
Belief policies and vectors

38 min worked math · 4 min exercise · 3 min recap

What changes from an MDP?

Uncertain motion and uncertain knowledge are different.

MDP

Motion may be random
Resulting state is observed
Policy uses state

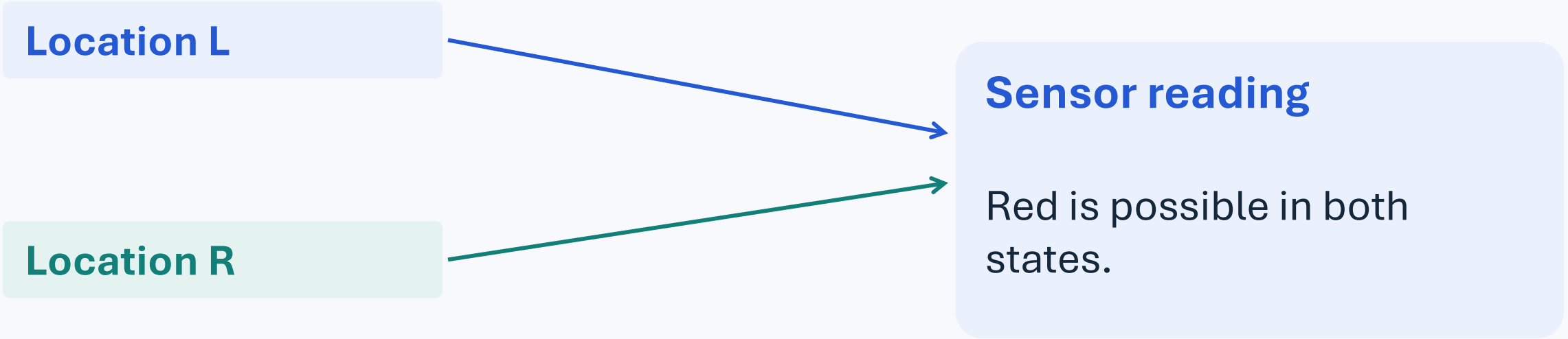
POMDP

True state is hidden
Receive partial evidence
Policy uses belief

Stochastic motion alone does not make a model a POMDP.

State versus observation

The physical situation is not the same as the sensor reading.



Two different locations can produce the same observation.

The POMDP model

Specify dynamics, evidence, and consequences.

$$\langle S, A, Tr, R, \Omega, O \rangle$$

State/action	Transition/reward	Evidence
S: hidden states	Tr: dynamics	Ω: observations
A: actions	R: rewards	O: likelihoods

Planning horizon h and discount γ are specified separately.

Fix the observation timing

Act first, transition, then receive evidence about the resulting state.

Old state s → **Action a** → **New state s'** → **Observe o**

$$\mathrm{Tr}(s,a,s') = P(S_{t+1}=s' \mid S_t=s, A_t=a)$$

$$O(s',a,o) = P(O_{t+1}=o \mid S_{t+1}=s', A_t=a)$$

Use the resulting state s' in the observation likelihood.

Why history matters

The same latest reading can follow different histories.

History 1

Earlier evidence favors L
Then a red reading

History 2

Earlier evidence favors R
Then the SAME red reading

Different prior knowledge → different posterior beliefs

Belief summarizes relevant history under the assumed model.

What is a belief?

A distribution over possible current states.

$$b_t(s) = P(S_t = s \mid h_t)$$

$$b_t(s) \geq 0; \quad \sum_s b_t(s) = 1$$

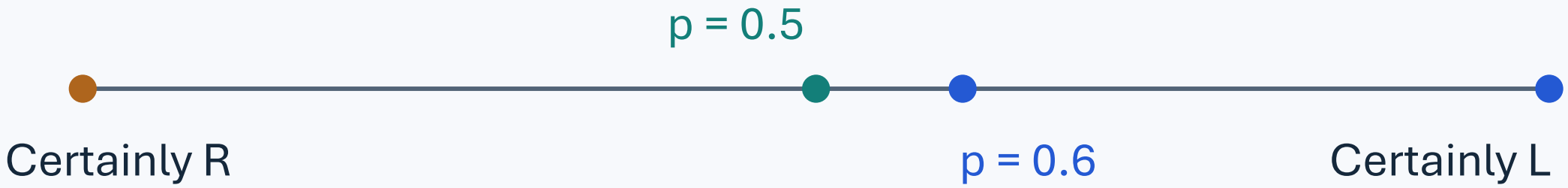
$b = (0.6, 0.4)$ is not the same as “certainly L”.

The most likely state alone discards decision-relevant uncertainty.

Belief space

Two hidden states already allow infinitely many beliefs.

$$b = (p, 1-p), \quad 0 \le p \le 1$$



m states: $\sum_i b_i = 1$ gives an $(m-1)$-dimensional simplex.

A point in belief space represents uncertainty, not a physical location.

Example A: move, then observe

TA-created model · prior b = (0.6, 0.4) in (L,R) order.

Move transition

From/to	L	R
L	0.8	0.2
R	0.1	0.9

Sensor model

State	Red	Blue
L	0.9	0.1
R	0.2	0.8

Transition rows: old state. Columns: resulting state.

Prediction by total probability

Average over the unknown old state.

$$\bar{b}(s') = \sum_s \mathrm{Tr}(s,a,s')\, b(s)$$

$$\bar{b}(\mathrm{L}) = 0.8 \times 0.6 + 0.1 \times 0.4 = 0.52$$

$$\bar{b}(\mathrm{R}) = 0.2 \times 0.6 + 0.9 \times 0.4 = 0.48$$

Prediction uses the action model, before the new observation.

Prediction in matrix form

Column beliefs and row-stochastic transitions require a transpose.

$$\bar{b} = T_a^T \, b$$

$$\begin{bmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{bmatrix} \times \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.52 \\ 0.48 \end{bmatrix}$$

Each destination collects probability mass from all old states.

Read the observation model

A likelihood answers: how likely is this reading in this state?

State	Red	Blue
L	0.9	0.1
R	0.2	0.8

Condition on state

Each row sums to 1.
A likelihood column need not.

P(red | L) is not P(L | red).

Derive the Bayesian correction

Conditional probability combines prediction with sensor evidence.

$$b^{ao}(s') = P(s' \mid b,a,o)$$

$$= P(o,s' \mid b,a) \,/\, P(o \mid b,a)$$

$$= O(s',a,o)\, \bar{b}(s') \,/\, P(o \mid b,a)$$

Likelihood × predicted probability, divided by observation probability.

Why normalize?

Weights are joint probabilities, not yet a posterior distribution.

$$w(s') = O(s', a, o)\, \bar{b}(s')$$

$$Z = P(o \mid b, a) = \sum_{s'} w(s')$$

$$b^{ao}(s') = w(s') / Z, \quad \text{for } Z > 0$$

Z = 0: the conditional posterior is undefined under the model.

Red observation: complete calculation

Start from the prediction (0.52, 0.48), not the original prior.

Stage	L	R	Sum
Prediction	0.52	0.48	1
Likelihood	0.9	0.2	—
Weights	0.468	0.096	0.564
Posterior	0.829787	0.170213	1

Red favors L, but does not make L certain.

Your turn: the blue alternative

Four minutes · use the SAME prediction $\bar{b} = (0.52, 0.48)$.

Stage	L	R	Sum
Prediction	0.52	0.48	1
Blue likelihood	0.1	0.8	—
Weights	?	?	?
Posterior	?	?	1

Find the weights, observation probability, and normalized belief.

Blue observation: answer and interpretation

A different observation leads to a different posterior.

Stage	L	R	Sum
Prediction	0.52	0.48	1
Likelihood	0.1	0.8	—
Weights	0.052	0.384	0.436
Posterior	0.119266	0.880734	1

Blue and red are alternatives, not consecutive corrections.

The complete belief update

Prediction and correction form one recursive estimator.

$$b^{ao}(s') = [\, O(s',a,o) \sum_s Tr(s,a,s') b(s) \,] \,/\, Z$$

$$Z = \sum_x O(x,a,o) \sum_s Tr(s,a,x) b(s)$$

$$\bar{b} = T_a^{\mathsf{T}}\, b; \quad w = D_{a,o}\, \bar{b}; \quad b^{ao} = w \,/\, (1^{\mathsf{T}}\, w)$$

D is diagonal: one likelihood per resulting state.

Check your update

Before planning, validate the probability calculation.

$$\sum_o P(o \mid b,a) = 0.564 + 0.436 = 1$$

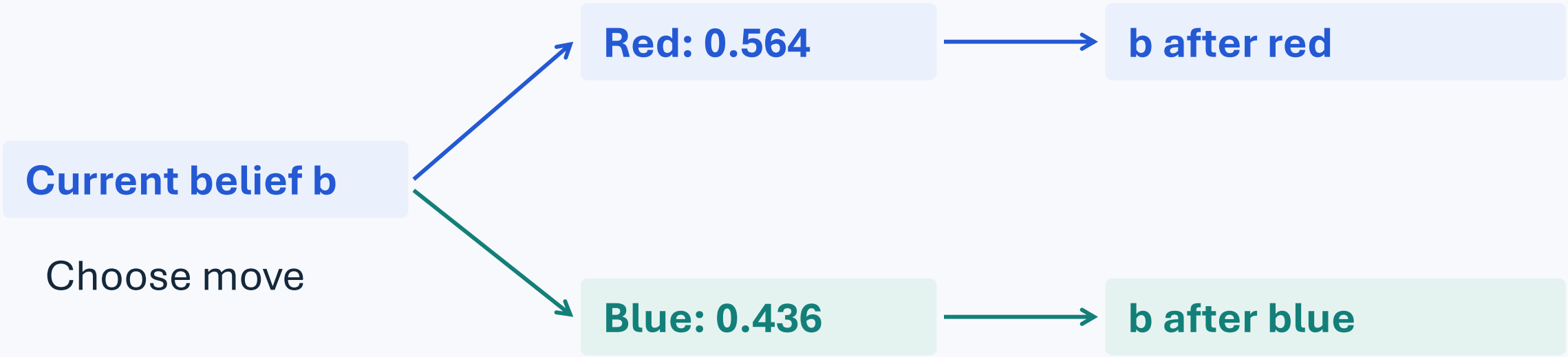
$$\sum_o P(o \mid b,a)\, b^{ao}(s') = \bar{b}(s')$$

$$0.564 \times 0.829787 + 0.436 \times 0.119266 \approx 0.52$$

Average the possible posteriors: recover the predicted belief.

An MDP over beliefs

The belief is observed by the agent; the physical state stays hidden.



Given b, a and o, the next belief is determined.

Policies and rewards over beliefs

Choose using uncertainty, not only the most likely state.

$a = \pi_h(b)$: policy for h decisions remaining

$$R(b,a) = \sum_s b(s)\, R(s,a)$$

Example: $b = (0.6,0.4)$, commit $L \rightarrow 0.6(10)+0.4(-10)=2$

Expected reward averages over the possible hidden states.

Bellman planning over beliefs

Choose an action, then average the possible observation branches.

$$V_0(b) = 0$$

$$V_h(b) = \max_a \left[R(b,a) + \gamma \sum_o P(o \mid b,a)\, V_{h-1}(b^{ao}) \right]$$

Maximum: action choice. Sum: uncertain observations. Update: future belief.

Example B: reset the problem

NEW question · prior b = (0.5,0.5) · no move before sensing.

Prior

Reset to (0.5,0.5)
No movement

Commit

Correct: +10
Incorrect: −10
Task ends

Sense

State unchanged
Pay cost 1
Then commit

Finite choice: commit now, or sense once and then commit. γ = 1.

Immediate commitment

Without additional evidence, both choices tie at the uniform prior.

$$V_L = 0.5 \times 10 + 0.5 \times (-10) = 0$$

$$V_R = 0.5 \times (-10) + 0.5 \times 10 = 0$$

$$V_{\text{commit}} = \max(0,0) = 0$$

Reach a decision now; the task then ends.

Sense and update

Identity transition: prediction equals the new uniform prior.

Red · probability 0.55

Posterior: (9/11, 2/11)
Choose L
Value: 70/11 ≈ 6.3636

Blue · probability 0.45

Posterior: (1/9, 8/9)
Choose R
Value: 70/9 ≈ 7.7778

After red choose L; after blue choose R.

Calculate the value of information

Pay once, then choose the commitment using the observation.

$$V_{\text{sense}} = -1 + 0.55(70/11) + 0.45(70/9)$$

$$= -1 + 3.5 + 3.5 = 6$$

$$P(\text{correct}) = 0.5(0.9) + 0.5(0.8) = 0.85$$

$$\text{Check: } 0.85(10) + 0.15(-10) - 1 = 6$$

Information is valuable because it changes the later decision.

When sensing is too expensive

Useful information is not automatically worth its cost.

$$\text{At } b = (0.5, 0.5): \quad V_{\text{sense}}(c) = 7 - c$$

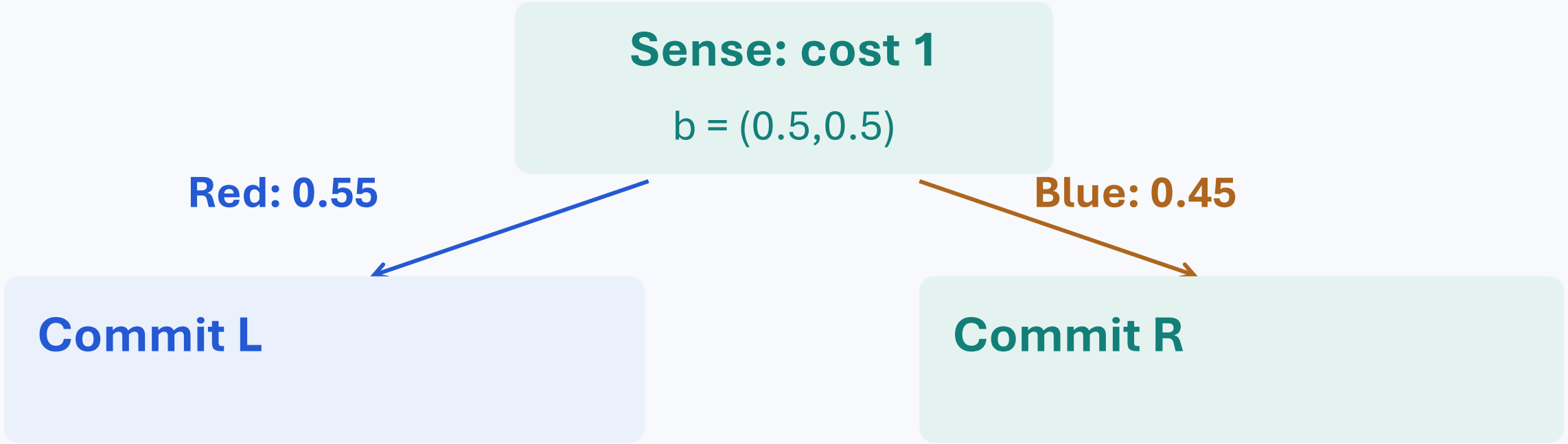
$$c < 7: \text{ sense wins}; \quad c = 7: \text{ tie}$$

$$c = 8: V_{\text{sense}} = -1 < V_{\text{commit}} = 0$$

This threshold depends on the prior, sensor, rewards, and horizon.

A contingent policy tree

The action after sensing depends on what was observed.



Actions are nodes; observations label the branches.

What does an alpha-vector store?

One fixed decision or tree, evaluated in each possible hidden state.

$$\alpha_L = (10, -10); \quad \alpha_R = (-10, 10)$$

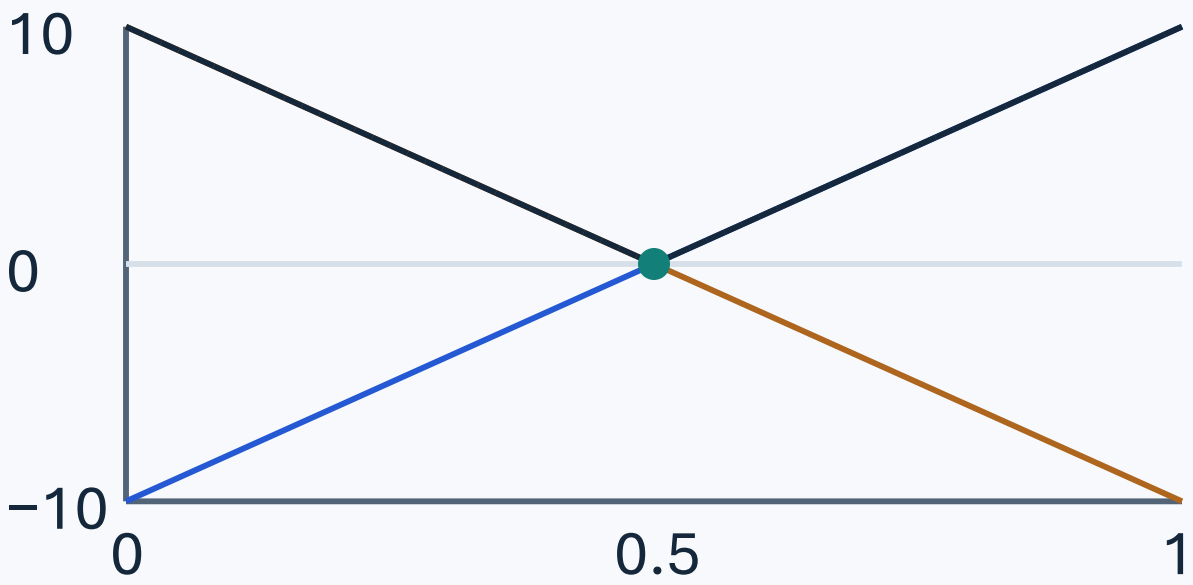
$$V_{\text{fixed}}(b) = b^T \alpha = \sum_s b(s)\alpha(s)$$

$\alpha(s)$: expected return of that fixed policy when starting in s

The vector stores values; it is not a belief distribution.

The commitment upper envelope

Only the two immediate commitments are included in this plot.



$$V_L(p) = 20p - 10$$

$$V_R(p) = 10 - 20p$$

$$V = \max(V_L, V_R)$$

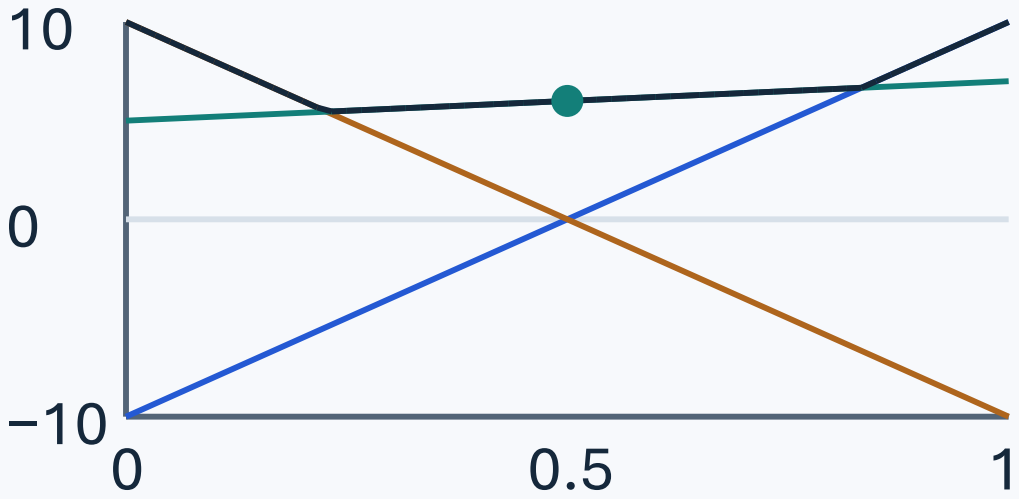
p = b(L)

Choose the larger value, rather than averaging the two policies.

The sensing tree also has a vector

Evaluate the same contingent tree separately from L and from R.

Add the fixed sensing tree



$$\alpha_{\text{sense}}(\text{L}) = -1 + 0.9(10) + 0.1(-10) = 7$$

$$\alpha_{\text{sense}}(\text{R}) = -1 + 0.2(-10) + 0.8(10) = 5$$

$$b^{\text{T}}\,\alpha_{\text{sense}} = 7p + 5(1-p) = 5 + 2p$$

At p = 0.5: the fixed sensing tree gives 6.

Why planning remains difficult

Compact beliefs do not make every POMDP easy to solve.

Belief space

Infinitely many possible beliefs

Policy trees

Branch on every future observation

Model limits

Our small example does not establish scalability.

Today: model → belief update → contingent decision → value representation.

Questions & recap

Reconstruct the update, then explain the decision.

Belief update

Prediction → likelihood → normalization

Information value

Does evidence change the best decision?

Policy values

Fixed tree → alpha-vector → belief dot product

Predict → weight → normalize → choose from the belief.

OPTIONAL / DEEPER DISCUSSION

Full conditioning and impossible evidence

The Bayesian update follows the chosen post-action model.

$$P(s',o \mid h_t,a) = O(s',a,o) \sum_s Tr(s,a,s')b_t(s)$$

$$P(s' \mid h_t,a,o) = P(s',o \mid h_t,a) \,/\, P(o \mid h_t,a)$$

$Z = 0$: every state with predicted mass gives likelihood zero.

Do not replace an undefined posterior with a fabricated distribution.

Informative rewards or other evidence belong in history and the model.

OPTIONAL / DEEPER DISCUSSION

Matrix conventions and posterior odds

Likelihood ratios quantify how evidence changes relative support.

Column belief: $\bar{b} = T_a^T b$; row belief: $\bar{b}^T = b^T T_a$

$D_{red} = \mathrm{diag}(0.9,0.2)$; $D_{blue} = \mathrm{diag}(0.1,0.8)$

$b^{ao}(L) / b^{ao}(R) = [O(L,a,o)/O(R,a,o)] \times [\bar{b}(L)/\bar{b}(R)]$

Red multiplier: 4.5; blue multiplier: 0.125

Odds formula requires nonzero denominator terms.

OPTIONAL / DEEPER DISCUSSION

General priors and sensing regions

Cost 1 · same sensor · commit now or sense once.

Red → choose L when $p \geq 2/11$.

Blue → choose R when $p \leq 8/9$.

Fixed tree (red L, blue R): $V = 5 + 2p$

Sense beats both immediate commitments for $5/22 < p < 5/6$.

At the region boundaries, sensing and a commitment tie.

OPTIONAL / DEEPER DISCUSSION

Policy-tree alpha-vector recursion

One root action and one fixed continuation tree per observation.

$$\alpha(s) = R(s,a)$$

$$+ \gamma \sum_{s'} \mathrm{Tr}(s,a,s') \sum_{o} O(s',a,o)\, \alpha_o(s')$$

Sense: $\mathrm{Tr} = I, \gamma = 1, R = -1$

$$\alpha = -1 + O(\mathrm{red}) \odot (10,-10) + O(\mathrm{blue}) \odot (-10,10) = (7,5)$$

α_o is the continuation vector selected for observation o.

OPTIONAL / DEEPER DISCUSSION

A map of solution approaches

Belief filtering and policy computation are separate tasks.

Exact planning

Alpha-vector candidates and pruning

Approximation

Selected beliefs / point-based methods

Online planning

Focus simulations on the current belief

No advanced solver or homework solution is implemented here.